

IEEE 830

Software Requirements Specification

Videoconference System (VCS)

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1. Introduction

1.1. Purpose

This document describes the software requirements and specifications for the videoconferencing software tailored for the professors at the Faculty of Computer Science and Engineering. The document is intended for the stakeholders and the developers. The software aims to facilitate online lectures, consultations, and conferences by offering features specifically designed to meet the needs of the educators, such as interactive whiteboards or advanced screen sharing capabilities.

The reader is assumed to have basic knowledge of utilizing videoconferencing software. Knowledge and understanding of UML diagrams is also required.

1.2. Scope

The web-based Videoconferencing system is designed to enable professors to hold online lectures, consultations, and conferences through a browser. The system will primarily use the university's Central Authentication System (CAS) for identifying the users, ensuring secure access for faculty members and students. Additionally, it will provide a registration page for new guest users, enabling them to register by providing an email address and creating a password, as well as allowing them to set their display name. The system allows existing guest users to login to their profiles. In addition to user authentication, the system will offer core features including:

- Advanced screen sharing capabilities.
- High-quality video and audio meetings.
- Interactive whiteboards for illustration in real-time.
- Chat functionality for an alternative mode of communication during sessions.

1.3. Definitions, Acronyms, Abbreviations

- **UML** – *Unified Modelling Language*, is a standardized modelling language consisting of an integrated set of diagrams, developed to help system and software developers for specifying, visualizing, constructing, and documenting the artifacts of software systems.
- **VCS** – *Videoconference System*, a system made for hosting virtual meetings, consultations, and video calls.
- **BBB** – *Big Blue Button*, is a virtual classroom software program designed for online education.
- **CAS** – *Central Authentication Service*, is a single sign-on protocol for the web that permits a user to access multiple applications while providing their credentials (such as user ID and password) only once.
- **EULA** – *End-user license agreement*, is a legal contract between a software supplier and a customer or end-user, generally made available to the customer via a retailer acting as an intermediary.
- **GDPR** – *General Data Protection Regulation* is a piece of European legislation that protects personal information. It outlines several requirements businesses must follow to process that data legally.
- **Cookie** (also known as a web cookie or browser cookie), is a small piece of data a server sends to a user's web browser.
- **Firewall** – is a network security device that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of security rules.
- **DDoS attack** – *denial-of-service attack*, is a cyber-attack in which the perpetrator seeks to make a machine or network resource unavailable to its intended users by temporarily or indefinitely disrupting services of a host connected to a network.
- **Multi-factor authentication** – is an electronic authentication method in which a user is granted access to a website or application only after successfully presenting two or more pieces of evidence to an authentication mechanism.

- **Password breach** – is when a cybercriminal has your password and is able to use it to get into your account.
- **Open port** – refers to a TCP or UDP port number that is configured to accept packets.

1.4. References

IEEE Standard 830-1998 for Requirements Specifications

Jose Montoya 2019, *SRS-Template*, <https://github.com/jam01/SRS-Template/blob/master/template.md#software-requirements-specification>

Michael Brown & Sheldon Linker 2021, *Appendix D: IEEE 830 Sample*, <https://press.rebus.community/requirementsengineering/back-matter/appendix-d-ieee-830-sample/>

1.5. Overview

The rest of this document is organized as follows. Section 2 provides information regarding the general description of the product, including constraints on the requirements, UML diagrams illustrating product features, assumptions about user knowledge, dependencies, and conditions under which the system may fail. Section 3 contains detailed information regarding the software requirements, addressing the functionality, stability, usability, performance, and accessibility of the software. Additionally, sample pages for the software interface are provided. The goals and feedback of the stakeholders is recorded in the validation section. The document concludes with an appendix of stakeholder interview notes.

2. General description

2.1. Product Perspective

2.1.1. Domain Model

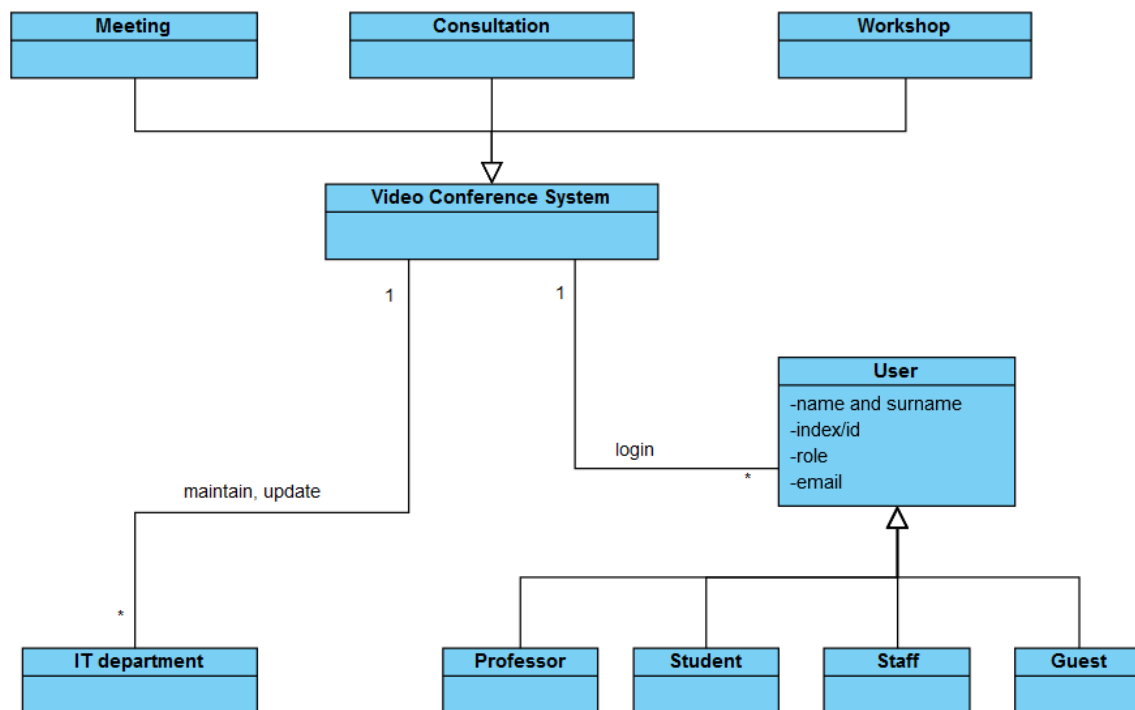
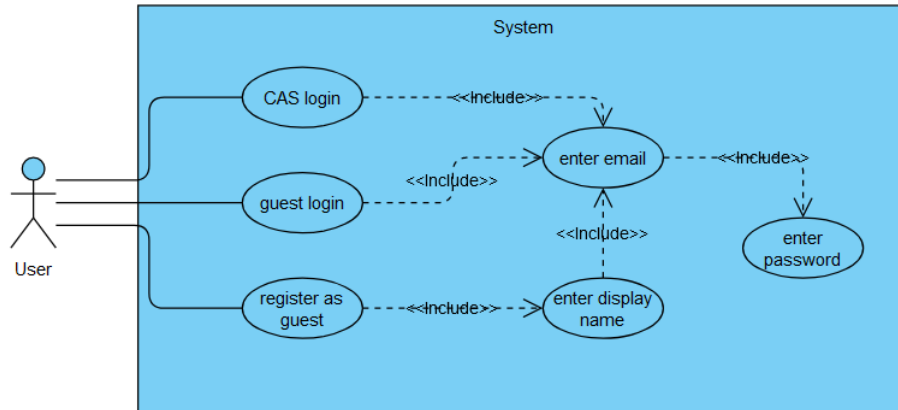


Figure 1 – Domain Model

The domain model in Figure 1 demonstrates the relationship between the users and the system. There are five types of users. Professors, students, members of the university's staff, along with guest members can log in to the system and use its features and services, while the members of the IT department maintain and update the system. The system offers services like meetings, consultations, and workshops.

2.2. Product Features

2.2.1. Login UML

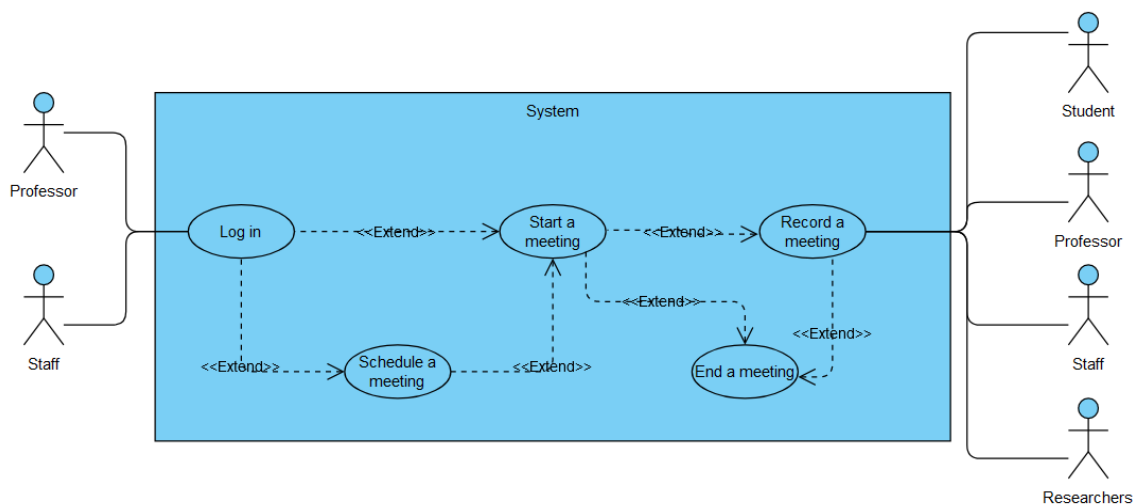


3.1.1, 3.1.2, 3.2.1.1

Figure 2 – Login Use case diagram

The use case diagram in Figure 2 shows that the users have three ways of logging into the system. The user can log in through the CAS system by entering their email and password. If the user is a new guest, they can register by entering a display name, a valid email address and password. If the user has an existing guest profile, they can log into it. After that they are logged in the system.

2.2.2. Meeting UML

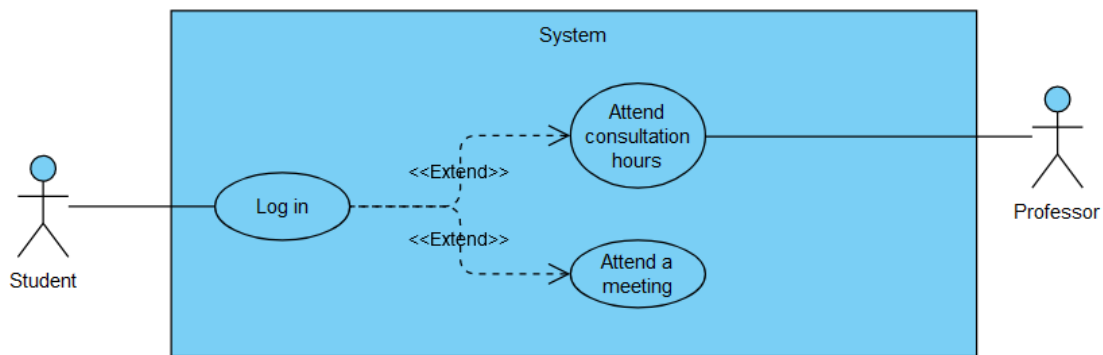


3.2.1.2

Figure 3 – Meeting Use case diagram

The use case in Figure 3 shows that after logging in, professors and staff members can either start or schedule a meeting. Additionally, they have the option to record the meeting. The recordings are available for all the authorized users. The professors and staff members have the option to fully end the meeting.

2.2.3. Student UML

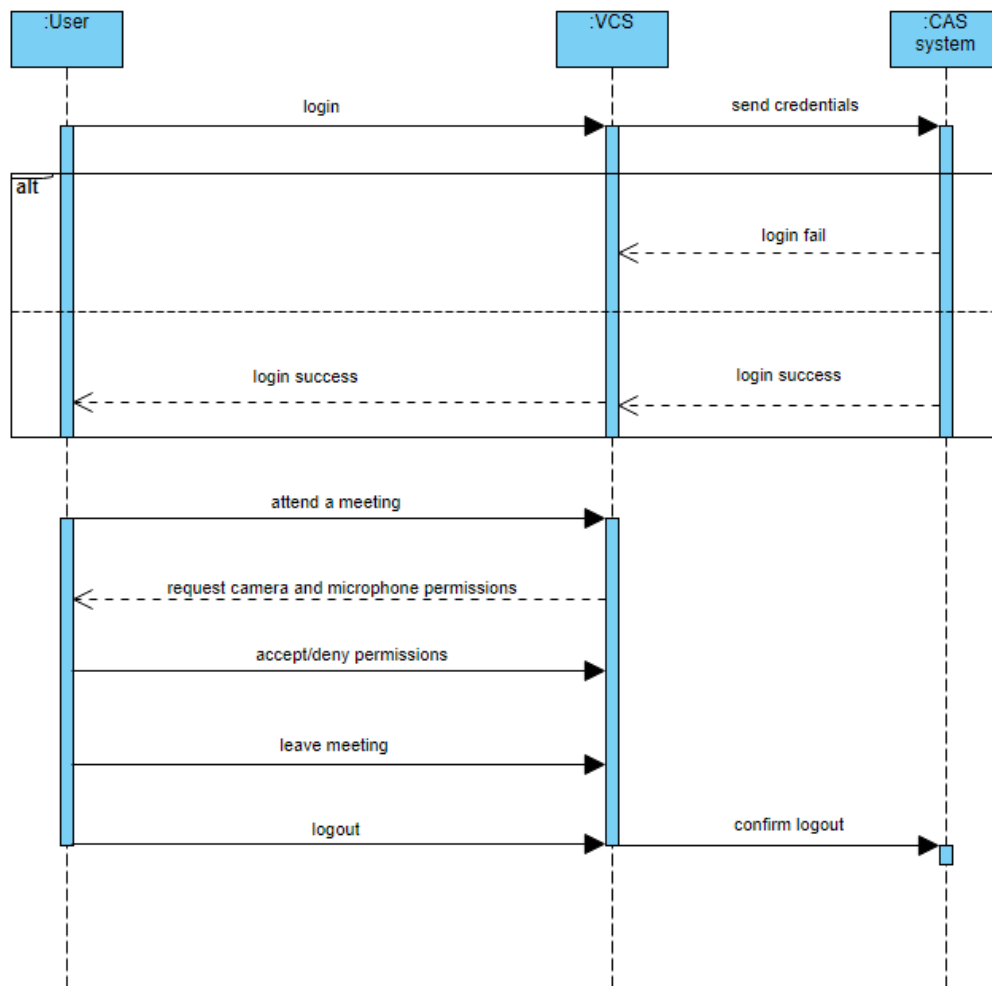


3.2.1.3

Figure 4 – Student Use case diagram

The use case in Figure 4 shows that after logging in the student has the options to attend a meeting and attend the professor's consultation hours.

2.2.4. Sequence Diagram

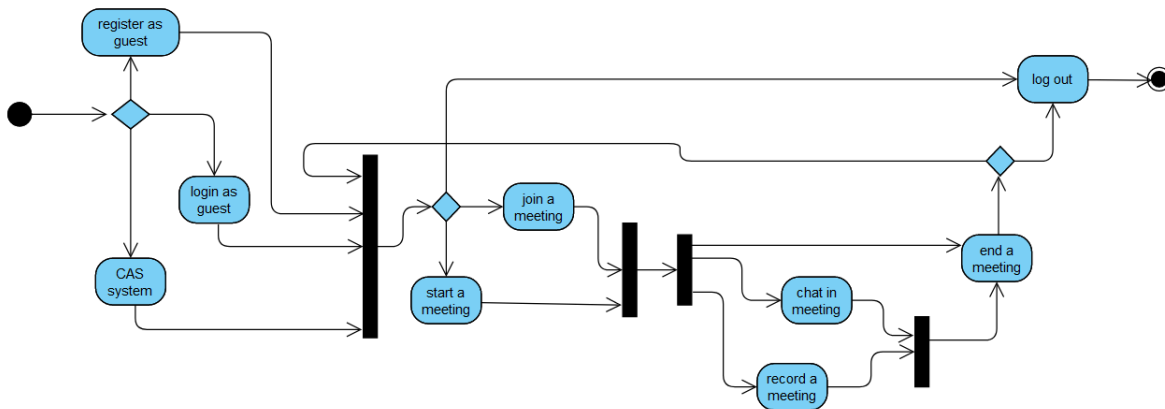


3.2.1

Figure 5 – Sequence diagram

The sequence diagram in Figure 5 illustrates the flow of actions users take sequentially. After the user successfully logs in the system, they attend a meeting, accept camera and microphone permissions, and leave the meeting when it ends. After the user is done using the system, they log out.

2.2.5. Activity Diagram



3.2.1

Figure 6 – Activity diagram

The activity diagram in Figure 6 displays the different actions the user can take while using the system. The user has three options for logging in, with the CAS system, with an existing guest account or registering as a guest. Then the user can either log out, join a meeting, or start a meeting. If they choose to join or start a meeting, they can chat with other users in the meeting, record the meeting or completely end the meeting. The user can decide to log out of the system or start a new meeting and go through the cycle of actions for meetings.

2.3. User Characteristics

There are several users of the VCS (Videoconference system); authorized and non-authorized users, as well as those who will maintain the system.

Authorized users can schedule, start, record and end meetings. Non-authorized users can attend meetings and chat with other users during the duration of them. The third type of users must be experienced network administrators that can update and maintain the system, as well as see all the information about other users registered to the system.

The authorized and non-authorized users are members of the general public that attend educational facilities.

2.4. General Constraints

- The hosts will need server infrastructure to host the service for users to access it.
- The user would need a device that supports internet connection along with a secure browser installed.
- The system shall follow the GDPR and EULA guidelines, as well as the data privacy laws within the country of use.
- The user would need a microphone to be able to transmit audio.
- The user would need a speaker for audio playback.
- The user would require a camera to enable video calls, allowing users to engage in face-to-face conversations.
- The hosts will need the infrastructure to facilitate up to 2000 concurrent users.

2.5. Assumptions and Dependencies

2.5.1. Assumptions

It is assumed that users have general knowledge of how to use a browser with an internet connection on their preferred device.

2.5.2. Dependencies

The system depends on the CAS system for user login.

2.5.3. Conditions for system failure

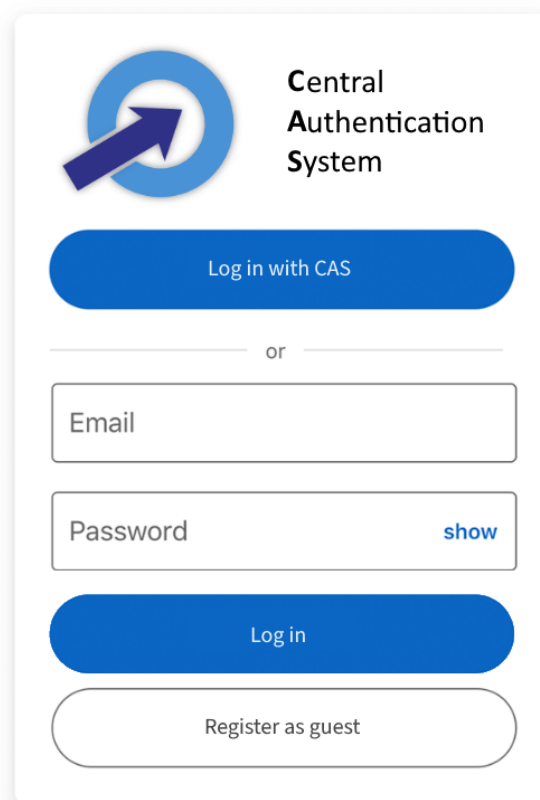
- *Power outage*: If the server infrastructure which hosts the system encounters a power outage, the system will be unavailable for all users.
- *Password breach*: Bad actors could gain access to the system by procuring login credentials of an admin user and could therefore disrupt the system.
- *DDoS attack*: Bad actors could overload the system using a distributed denial of service attack to restrict access to the system or even temporarily disable the system.
- *Open port attack*: The system requires open ports to function, thus if they are not secured correctly, it could enable bad actors to access the system directly.

- *CAS failure*: If the CAS system fails, the users won't be able to log in the system.

3. Specific requirements

3.1. External interfaces

3.1.1. “Login” Sample Page

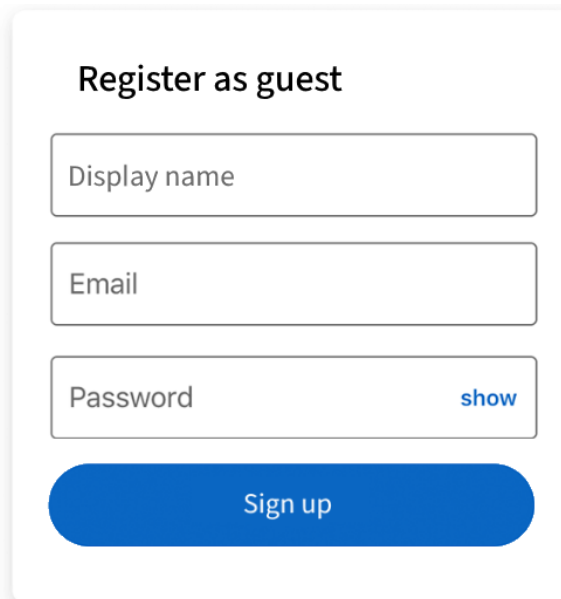


The login page features a central white card with a light gray border. At the top left is a logo consisting of a blue circle with a white arrow pointing clockwise. To the right of the logo, the text 'Central Authentication System' is displayed in a bold, black, sans-serif font. Below the logo and text is a large, rounded blue button with the text 'Log in with CAS' in white. Underneath this button is a horizontal line with the word 'or' centered. Below the line are two input fields: the first is labeled 'Email' and the second is labeled 'Password'. The 'Password' field has a small blue 'show' link to its right. Below the input fields is a large, rounded blue button with the text 'Log in' in white. At the bottom of the card is a rounded white button with a gray border and the text 'Register as guest' in gray.

3.2.1.1

Figure 7 – Login page

3.1.2. “Register” Sample Page



A registration form titled "Register as guest". It contains three input fields: "Display name", "Email", and "Password". The "Password" field has a "show" link to its right. Below the fields is a blue "Sign up" button.

Register as guest

Display name

Email

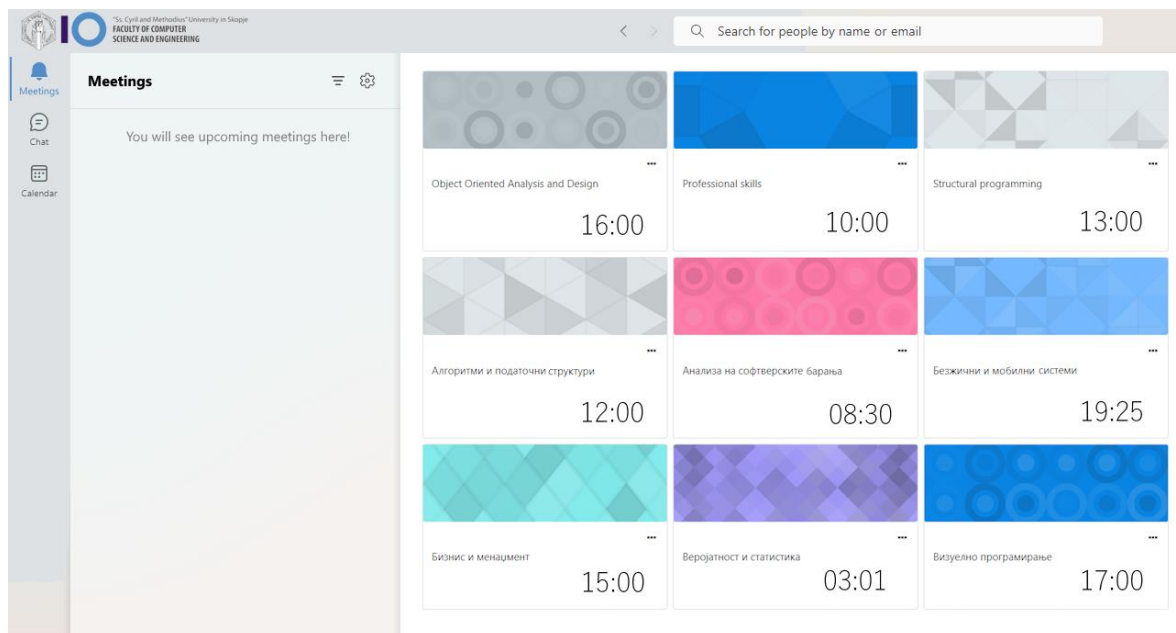
Password [show](#)

Sign up

3.2.1.1

Figure 8 – Register page

3.1.3. “Home” Sample Page



The home page interface for the Faculty of Computer Science and Engineering at "St. Cyril and Methodius" University in Skopje. It features a sidebar with "Meetings", "Chat", and "Calendar" options. The main area displays a grid of meeting cards with titles in English and Macedonian, and times.

St. Cyril and Methodius University in Skopje
FACULTY OF COMPUTER SCIENCE AND ENGINEERING

Search for people by name or email

Meetings

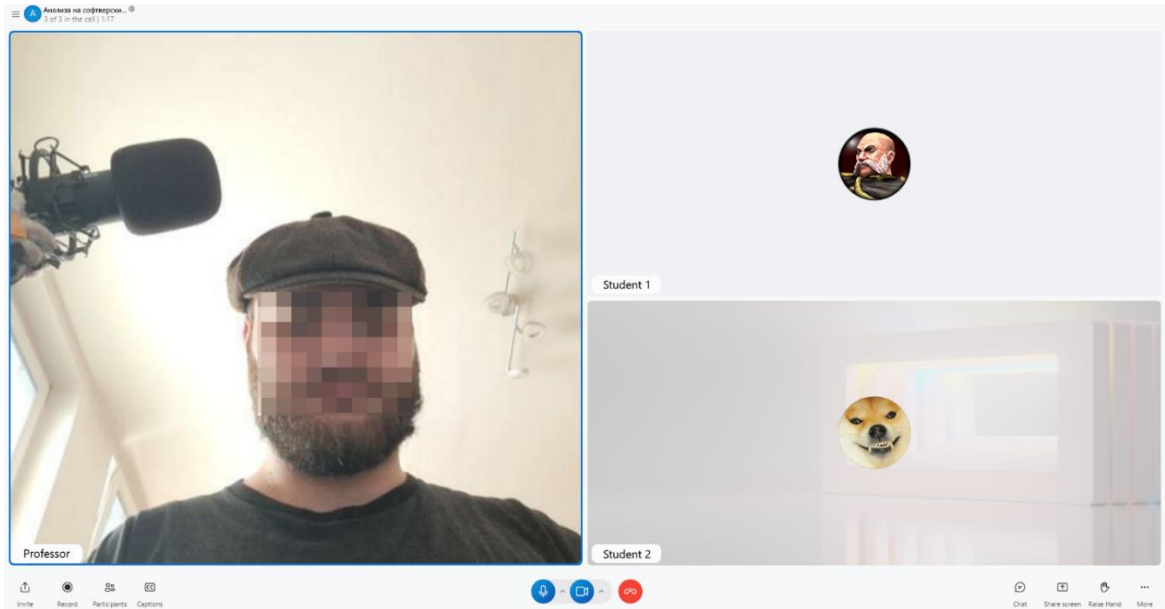
You will see upcoming meetings here!

Object Oriented Analysis and Design 16:00	Professional skills 10:00	Structural programming 13:00
Алгоритми и подложни структури 12:00	Анализа на софтверските барања 08:30	Безжични и мобилни системи 19:25
Бизнис и менаџмент 15:00	Веројатност и статистика 03:01	Визуелно програмирање 17:00

3.2.1.4

Figure 9 – Home page

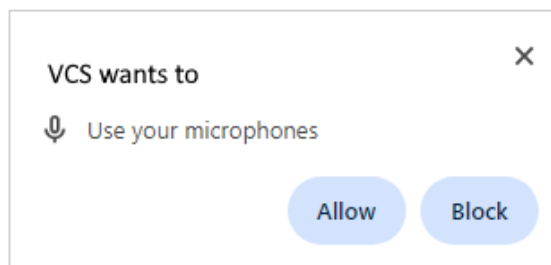
3.1.4. “Meeting” Sample Page



3.2.1.5

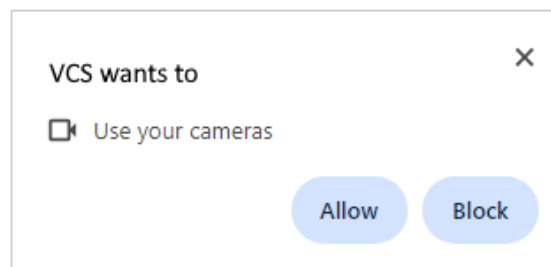
Figure 10 – Meeting page

3.1.5. Permissions prompts



3.2.1.5.1

Figure 11 – Microphone permissions



3.2.1.5.2

Figure 12 – Camera permissions

3.1.6. Cookies permission

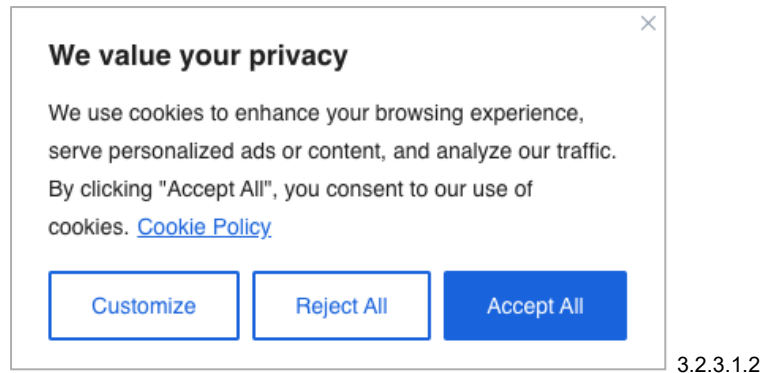


Figure 13 – Cookies permission

3.1.7. EULA acceptance

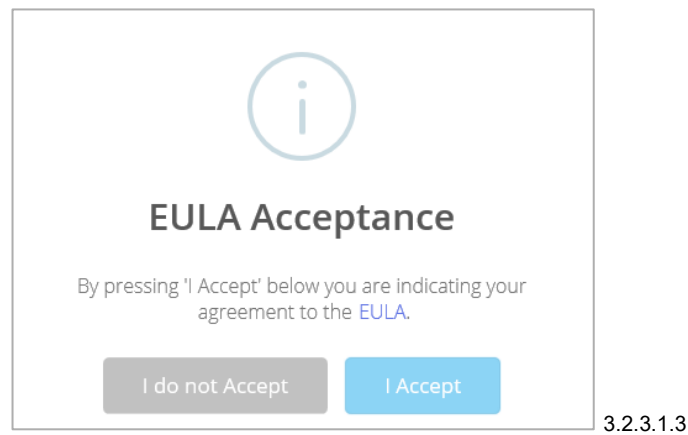


Figure 14 – EULA acceptance

3.2. Functional requirements

Priority level	Description
Priority 1	Necessary for system functionality
Priority 2	Benefitable functionality
Priority 3	Other features

Figure 15 – Priority requirements table

3.2.1. Functionality

3.2.1.1. User login

- 3.2.1.1.1. The system shall allow the user to log in with CAS. *Priority 1*
- 3.2.1.1.2. The system shall include a login functionality for existing guest users. *Priority 1*
- 3.2.1.1.3. The existing guest user shall enter their email and password to login. *Priority 1*
- 3.2.1.1.4. The system shall allow new users to register. *Priority 1*
- 3.2.1.1.5. The new user shall enter a display name. *Priority 1*
- 3.2.1.1.6. The new user shall enter a valid email address. *Priority 1*
- 3.2.1.1.7. The new user shall enter a password. *Priority 1*

3.2.1.2. Authorized user functions for meetings

- 3.2.1.2.1. The system shall allow the authorized users to start a meeting. *Priority 1*
- 3.2.1.2.2. The system shall allow the authorized user to schedule a meeting. *Priority 2*
- 3.2.1.2.3. The system shall allow the authorized users to record the meeting. *Priority 2*
- 3.2.1.2.4. The authorized users shall have access to the recorded meeting. *Priority 1*
- 3.2.1.2.5. The system shall allow authorized users to download the meeting recordings. *Priority 1*
- 3.2.1.2.6. The system shall allow authorized users to invite other users to the meeting. *Priority 1*
- 3.2.1.2.7. The system shall allow the authorized users to remove participants from the meeting. *Priority 1*
- 3.2.1.2.8. The system shall allow authorized users to have consultation hours. *Priority 2*

3.2.1.3. *Non-authorized user functions*

- 3.2.1.3.1. The system shall allow non-authorized users to attend meetings with permission. *Priority 1*
- 3.2.1.3.2. The system shall allow non-authorized users to attend consultation hours with permission. *Priority 1*

3.2.1.4. *Home page*

- 3.2.1.4.1. The system shall display a list of upcoming daily meetings. *Priority 1*
- 3.2.1.4.2. The system shall display the name and starting time of the upcoming meeting. *Priority 1*
- 3.2.1.4.3. The system shall allow the user to search other users by name or email. *Priority 2*

3.2.1.5. *Meeting settings*

- 3.2.1.5.1. The system shall ask the user for microphone permissions. *Priority 1*
- 3.2.1.5.2. The system shall ask the user for camera permissions. *Priority 1*
- 3.2.1.5.3. The system shall allow the user to mute themselves. *Priority 1*
- 3.2.1.5.4. The system shall have time stamp messages in the meeting chat. *Priority 1*
- 3.2.1.5.5. The system shall allow users to share their screen during a meeting. *Priority 1*
- 3.2.1.5.6. The system shall allow the users to change the volume during the meeting. *Priority 1*
- 3.2.1.5.7. The system shall allow the users to mute the meeting. *Priority 1*
- 3.2.1.5.8. The system shall allow users to attach files. *Priority 2*
- 3.2.1.5.9. The system shall allow playback of the recorded meetings. *Priority 2*
- 3.2.1.5.10. The system shall allow the user to change the speed on the meeting playbacks. *Priority 3*
- 3.2.1.5.11. The system shall allow the user to change their background when the camera is in use. *Priority 3*

- 3.2.1.5.12. The system shall have break points on the meeting playbacks. *Priority 3*
- 3.2.1.5.13. The system shall allow the user to change the resolution on meeting playbacks. *Priority 3*
- 3.2.1.5.14. The system shall allow the authorized users to pin messages in the meeting chat. *Priority 3*
- 3.2.1.5.15. The system shall notify the users ten minutes before a meeting starts. *Priority 1*
- 3.2.1.5.16. The system shall have full screen mode for meetings. *Priority 3*
- 3.2.1.5.17. The system shall have full screen mode on meeting playbacks. *Priority 3*

3.2.1.6. *Calendar*

- 3.2.1.6.1. The system shall have personal user calendars. *Priority 3*
- 3.2.1.6.2. The system shall have group calendars. *Priority 3*
- 3.2.1.6.3. The system shall allow the authorized user to schedule events on the group calendar. *Priority 3*

3.2.1.7. *Whiteboard*

- 3.2.1.7.1. The system shall have a whiteboard. *Priority 2*

3.2.2. *Usability*

3.2.2.1. *Graphical User Interface*

- 3.2.2.1.1. The system shall allow the users to change their profile picture. *Priority 3*

3.2.2.2. *Accessibility*

- 3.2.2.2.1. The system shall have text to speech for visually impaired users. *Priority 2*
- 3.2.2.2.2. The system shall have auto-generated captions on meetings for hearing impaired users. *Priority 2*
- 3.2.2.2.3. The system shall have multi-language support. *Priority 2*

3.2.3. Security

- 3.2.3.1.1. The system shall use the GDPR cookie policy. *Priority 1*
- 3.2.3.1.2. The system shall ask the users for permission to use cookies. *Priority 1*
- 3.2.3.1.3. The system shall follow the EULA policy. *Priority 1*
- 3.2.3.1.4. The system shall use a multi-factor authentication method when a security breach occurs. *Priority 1*
- 3.2.3.1.5. The system shall have a Firewall. *Priority 1*

3.3. Performance requirements

- 3.3.1. The system shall operate with 99.99% uptime in a week. *Priority 1*
- 3.3.2. The system shall withstand a maximum of 2000 users. *Priority 1*
- 3.3.3. The system shall have a maximum resolution of 1080p for the meetings. *Priority 1*

4. Validation

Three stakeholders are involved in the creation of the Videoconferencing system: a professor, the head of the computer center at the Faculty of Computer Science and Engineering, and the head of the faculty's legal department. Each stakeholder has specific goals for the functionality of the system.

The professors at the faculty require a web-based platform which shall offer them the opportunity to communicate with students, other professors, staff, and guest lecturers through video meetings. The system shall only grant certain permissions to the professors regarding starting and recording meetings, along with inviting users to attend them.

The head of the faculty's legal department states that system must comply with applicable privacy laws and implement measures to ensure the security of user data.

The head of the computer center at the faculty necessitates that the system is capable to undergo regular maintenance and updates to ensure its functionality. Furthermore, the system shall be protected from unauthorized access by non-affiliated parties.

This SRS document correctly describes the intended system capabilities and characteristics that shall satisfy the unique requirements of the stakeholders. This provides an appropriate basis to proceed with the development of the software.

5. Appendices

5.1. Stakeholder interview notes

- web based conference rooms
- sessions cannot be done with BBB
- consultations, workshops
- 99,99% or 99,95% in a week
- scalable (used outside of the faculty)
- lectures should mimic real life
- firewall - features dropping out suspicious connections
- max 2000 people
- high quality video
- automated updates
- it should have/ask for cookies
- only professors and the IT department can have access to the recordings
- download recordings
- everything to be done by the laws in the country
- law for privacy of users
- EULA should be applied